Denoising Diffusion Probabilistic Model for Channel Estimation

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Introduction

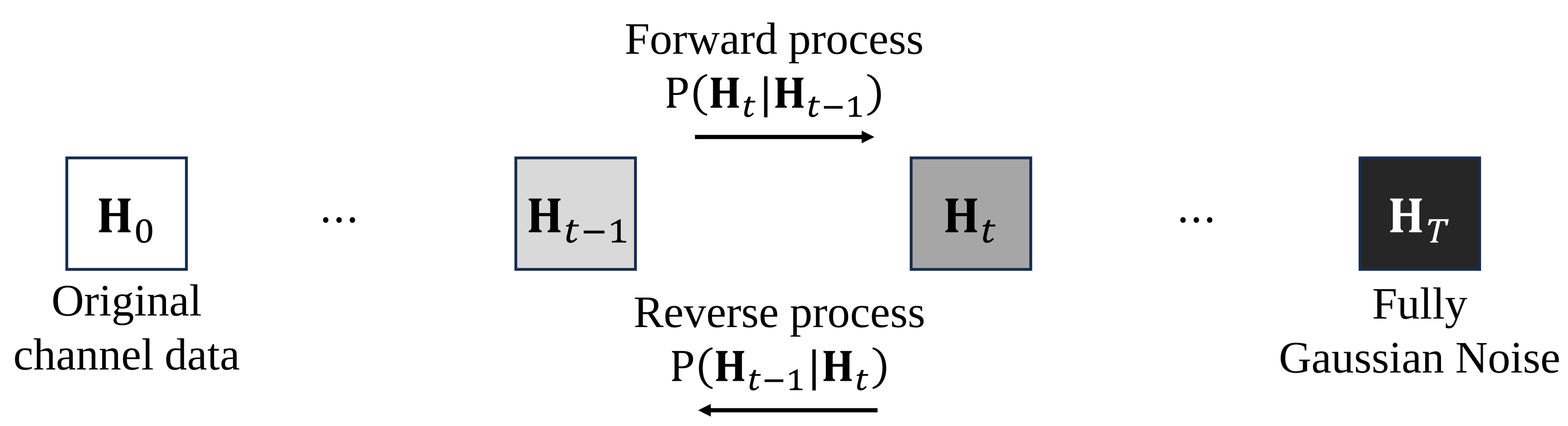
Motivation

- Traditional estimation techniques often struggle to maintain accuracy in low SNR scenarios resulting in compromised performance.
- We aimed to improve the reliability and precision of channel estimation by leveraging generative AI technologies.

Contribution

- DDPM[1][2] for channel estimation enhanced estimation performance than traditional methods in all SNR ranges.
- The DDPM for channel estimation incrementally learns the noise distribution from noisy data, allowing for more effective denoising and improved accuracy in complex scenarios.

Denoising Diffusion Probabilistic Model[2]



Forward process

- The forward process refers to the sequence of steps in a diffusion model where noise is progressively added to the channel data over a series of diffusion steps.

$$\mathrm{P}(\mathbf{H}_t|\mathbf{H}_{t-1}) \cong \mathcal{CN}(\mathbf{H}_t; \sqrt{1-\beta_t}\mathbf{H}_{t-1}, \beta_t \mathbf{I}) \quad (1)$$

$$\mathrm{P}(\mathbf{H}_t|\mathbf{H}_0) \cong \mathcal{CN}(\mathbf{H}_t; \sqrt{\bar{\alpha}_t}\mathbf{H}_0, (1-\bar{\alpha}_t)\mathbf{I}) \quad (2)$$

› β_t: diffusion scheduler[5, Table 1]
› $\alpha_t \cong 1-\beta_t,\ \ \bar{\alpha}_t \cong \prod_{k=1}^{t} \alpha_k$

- $\mathbf{H}_t$ can expressed using $\mathbf{H}_0$ and $\boldsymbol{\epsilon} \sim \mathcal{CN}(0, \mathbf{I})$.

$$\mathbf{H}_t(\mathbf{H}_0, \boldsymbol{\epsilon}) = \sqrt{\bar{\alpha}_t}\mathbf{H}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon} \quad (3)$$

Reverse process

- The reverse process refers to the sequence of steps where the model learns to reverse the diffusion forward process, effectively reconstructing the original channel data from noisy diffusion steps.

$$\mathrm{P}(\mathbf{H}_{t-1}|\mathbf{H}_t) \cong \mathcal{CN}(\mathbf{H}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{H}_t, t), \sigma_t^2 \mathbf{I}) \quad (4)$$

› $\boldsymbol{\mu}_\theta(\mathbf{H}_t, t)$: Neural Net model to estimate mean of $\mathrm{P}(\mathbf{H}_{t-1}|\mathbf{H}_t)$
› σ_t^2: variance of $\mathrm{P}(\mathbf{H}_{t-1}|\mathbf{H}_t)$, dependent by diffusion step t

Training

- (4) can be expressed as a closed form conditional probability distribution by applying the Rao-Blackwellized theorem[2, Section 3.1].

$$\mathrm{P}(\mathbf{H}_{t-1}|\mathbf{H}_t, \mathbf{H}_0) \cong \mathcal{CN}(\mathbf{H}_{t-1}; \tilde{\boldsymbol{\mu}}_t(\mathbf{H}_t, \mathbf{H}_0), \tilde{\beta}_t \mathbf{I}) \quad (5)$$

› $\tilde{\beta}_t \cong \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$

- $\tilde{\boldsymbol{\mu}}_t(\mathbf{H}_t, \mathbf{H}_0)$ is utilized to target during $\boldsymbol{\mu}_\theta(\mathbf{H}_t, t)$ training.

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{H}_t, \mathbf{H}_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{H}_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{H}_t \quad (6)$$

- So, $\boldsymbol{\mu}_\theta(\mathbf{H}_t, t)$ can formulate using (5)

$$\begin{aligned}\boldsymbol{\mu}_\theta(\mathbf{H}_t, t) &= \tilde{\boldsymbol{\mu}}_t\left(\mathbf{H}_t, \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{H}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_\theta(\mathbf{H}_t, t)\right)\right) \\ &= \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{H}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{H}_t, t)\right)\end{aligned} \quad (7)$$

- Ultimately, DDPM predicts $\boldsymbol{\epsilon}$ conditioned on $\mathbf{H}_t$, i.e. $\boldsymbol{\epsilon}_\theta(\mathbf{H}_t, t)$.

$$\text{Loss function}(\theta) \coloneqq \mathbb{E}_{t,\mathbf{H}_0,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{H}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t\right)\right\|^2\right] \quad (8)$$

System Model & Channel Estimation

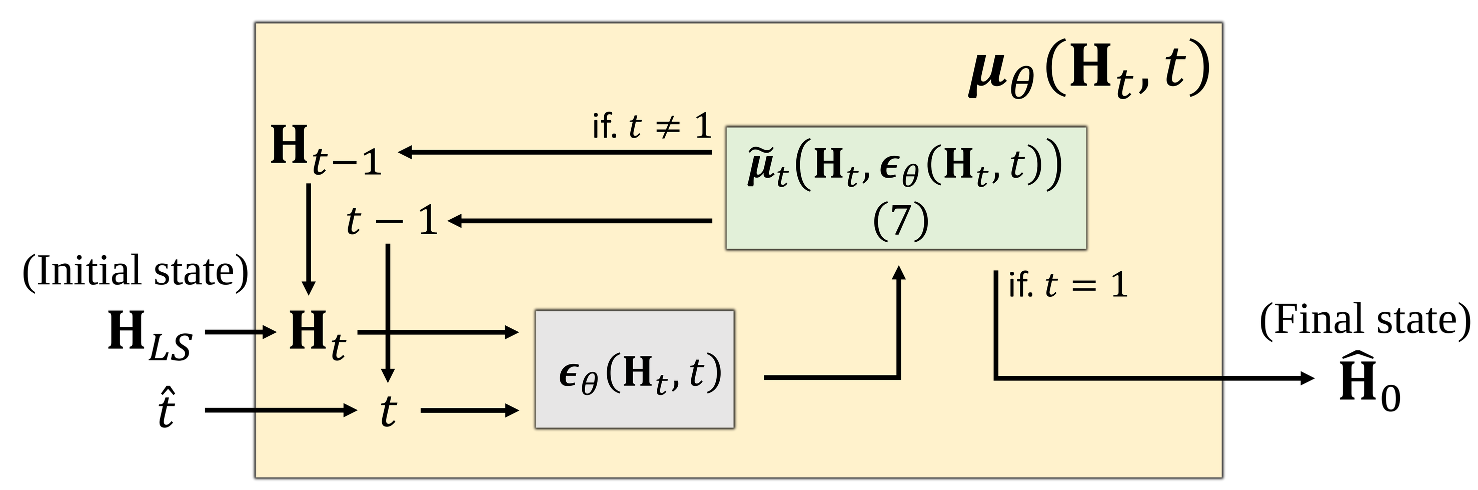
MIMO system model

$$\boldsymbol{y} = \mathbf{H}\mathbf{P} + \boldsymbol{N} \in \mathbb{C}^{N_{\mathrm{rx}} \times N_{\mathrm{P}}} \quad (9)$$

$$\mathbf{H}_{LS} = \boldsymbol{y}\mathbf{P}^{\boldsymbol{H}} = \mathbf{H} + \boldsymbol{N}\mathbf{P}^{\boldsymbol{H}} \quad (10)$$

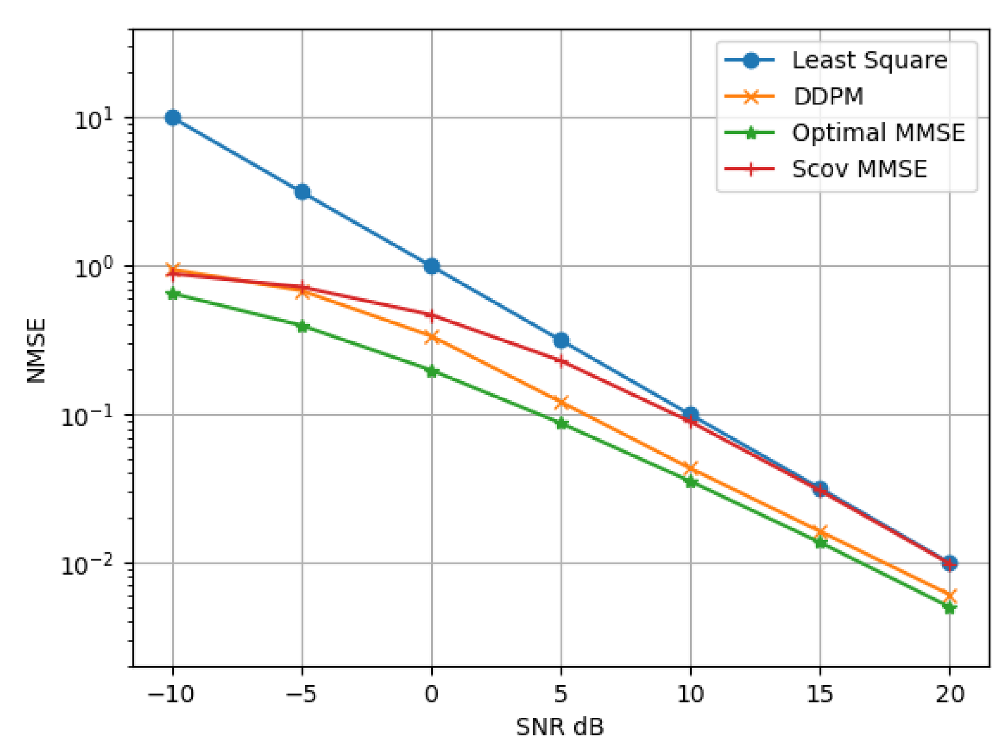
› $\mathbf{H} \in \mathbb{C}^{N_{\mathrm{rx}} \times N_{\mathrm{tx}}}$: MIMO channel matrix, $\mathbf{H} \sim \mathcal{CN}(0, \boldsymbol{C}_\delta)$
› $\mathbf{P} \in \mathbb{C}^{N_{\mathrm{tx}} \times N_{\mathrm{P}}}$: pilot matrix, $N_{\mathrm{tx}} = N_{\mathrm{P}}$ normalized Hadamard matrix
› $\boldsymbol{y}$: Rx received signal
› $N_{\mathrm{rx}}, N_{\mathrm{tx}}, N_{\mathrm{P}}$: the number of receive, transmitter antennas and pilots
› $\boldsymbol{N} \sim \mathcal{CN}(0, \mathbf{I})$: Additional White Gaussian Noise(AWGN)

- The covariance matrix $\boldsymbol{C}_\delta = \boldsymbol{C}_{\mathrm{rx}} \otimes \boldsymbol{C}_{\mathrm{tx}}$, $\otimes$ is Kronecker product. $\boldsymbol{C}_{\mathrm{tx}}, \boldsymbol{C}_{\mathrm{rx}}$ are channel covariance matrices, as defined in 3GPP cellular networks[3][4].
- The channel matrix $\mathbf{H}$ is transformed into angular domain using the FFT ,which serves as the input $\mathbf{H}_0$ for the aforementioned DDPM.
- DDPM model $\boldsymbol{\epsilon}_\theta$ sharing parameter θ across all diffusion steps.

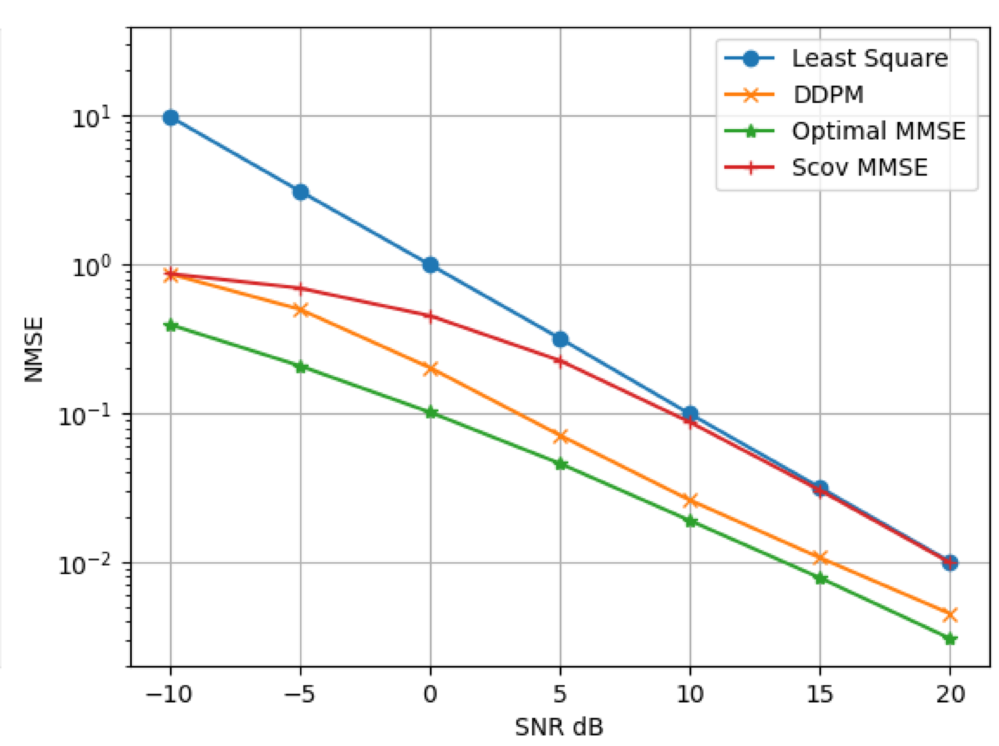


- DDPM based estimator achieves convergence to the ground truth Conditional Mean Estimator(CME), as the number of diffusion steps increases[5, Section 4.3].
- In test, DDPM initializes $\mathbf{H}_t$ to $\mathbf{H}_{LS}$ and uses it as input to the model. $\hat{t}$ is SNR level of $\mathbf{H}_{LS}$. Finally, $\widehat{\mathbf{H}}_0 = \boldsymbol{\mu}_\theta(\boldsymbol{\mu}_\theta(\cdots\boldsymbol{\mu}_\theta(\mathbf{H}_t, t)\cdots))$.

Numerical Result



$N_{\mathrm{rx}} = 8, N_{\mathrm{tx}} = 1$
SIMO Channel Estimation



$N_{\mathrm{rx}} = 8, N_{\mathrm{tx}} = 4$
MIMO Channel Estimation

- Numerical results compare the performance of channel estimation performance between the traditional channel estimation and the DDPM, using Normalized Mean Square Error(NMSE).
- The Optimal MMSE used true channel covariance $\boldsymbol{C}_\delta$ and Scov MMSE used sample channel covariance about training channel dataset[6].
- The proposed DDPM channel estimator uses 100 diffusion steps($T = 100$), with setting $\alpha_1 = 0.9999, \alpha_T = 0.9$. These settings enable training across diffusion stages ranging from 40dB to -22dB.

Reference

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